EKURHULENI WEST TVET COLLEGE

Surname & Initials	
ID Number	
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Facilitator	**RAPHATELO MD**
Programme/Course	**ICT**
Module	**COMPUTER PROGRAMMING L2**
Module Code	**CPROGL2**
Assessment Type	**PRACTICAL FORMATIVE ASSESSMENT**
Topic	**Customer Shopping Bill – Python**
Instructions	**1. Answer ALL questions.** **2. Use Python 3 in Thonny.** **3. Submit your IPO chart, Python file and test record.**
Duration	**60 minutes**
Date	

For Office Use ONLY		
Question No.	Marker	Moderator
1		
2		
3		
4		
5		
6		
7		
Raw Mark /25		Percentage:

Question 1

A small stationery shop needs a Python program to calculate the amount a customer must pay for notebooks. Customers receive a 10% discount only when the subtotal is R50.00 or more; otherwise no discount applies.

The program must ask for the customer name, quantity of notebooks and price per notebook in rand. It must calculate the subtotal, discount and final amount, and display all monetary values to two decimal places. Assume that all inputs are valid and quantity is at least one.

1.1. Draw an IPO chart using the above scenario. (4)

1.2. Capture the customer name, quantity and unit price using meaningful variable names, prompts showing units where applicable, and correct int/float conversions. (5)

1.3. Calculate the subtotal and use an if/else statement to apply the following discount conditions: (4)
- If the subtotal is R50.00 or more, calculate a 10% discount.
- Otherwise, set the discount to R0.00.

Calculate the final amount after applying the discount, using appropriate variables.

1.4. Display the customer name, subtotal, discount and final amount using clear labels, appropriate text case, rand units and two decimal places. (4)

1.5. Add a meaningful header comment describing the purpose of your program. (1)

1.6. Run all three test cases, ensuring that both conditions of the if/else statement are tested. Record the expected and actual final amounts and indicate whether the results match. (2)

Test	Customer	Quantity	Price (R)	Subtotal (R)	Discount (R)	Final (R)
1	Lebo	5	20.00	100.00	10.00	90.00
2	Thabo	2	15.50	31.00	0.00	31.00
3	Naledi	2	25.00	50.00	5.00	45.00

Actual results and comparison:

Test	Actual final amount (R)	Matches expected? (Yes/No)
1		
2		
3		

Question 2

Study these two separate examples of incorrect Python code:

```
Example A – Type error:
quantity = input("Enter quantity: ")
price = input("Enter price (R): ")
subtotal = quantity * price
Example B – Operator precedence: result = 10 + 5 * 2
```

2.1 Explain the type of error in Example A and correct it. (2)

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2.2 What value does Example B calculate, and why? (2)

_____

_____

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2.3 Rewrite Example B so that addition happens first and the result is 30. (1)

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TOTAL: 25 MARKS